

Telecom Customer Churn Prediction Using Machine Learning

Global Consumer Intelligence (GCI) Final Assignment

Submitted by

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Business Problem

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Telecommunication companies face significant financial losses due to customer churn. Identifying customers who are likely to leave enables the company to take proactive actions, improve customer satisfaction, and increase long-term revenue.

- Predict customer churn
- Reduce revenue loss
- Support retention campaigns

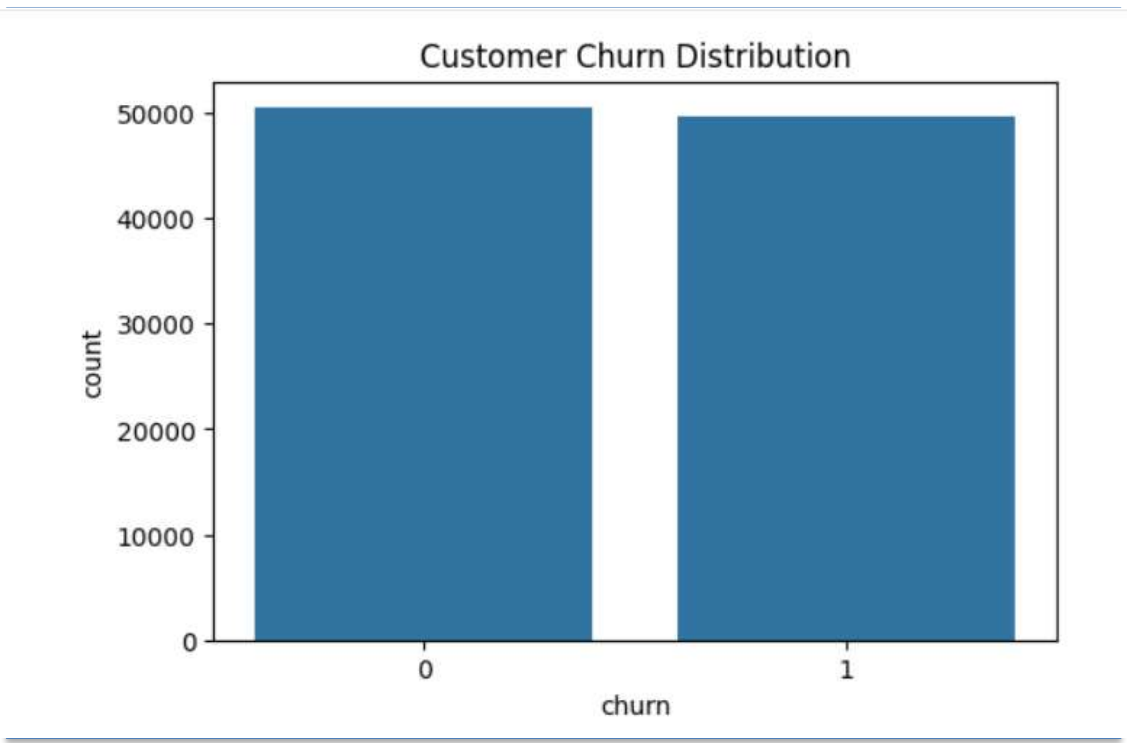
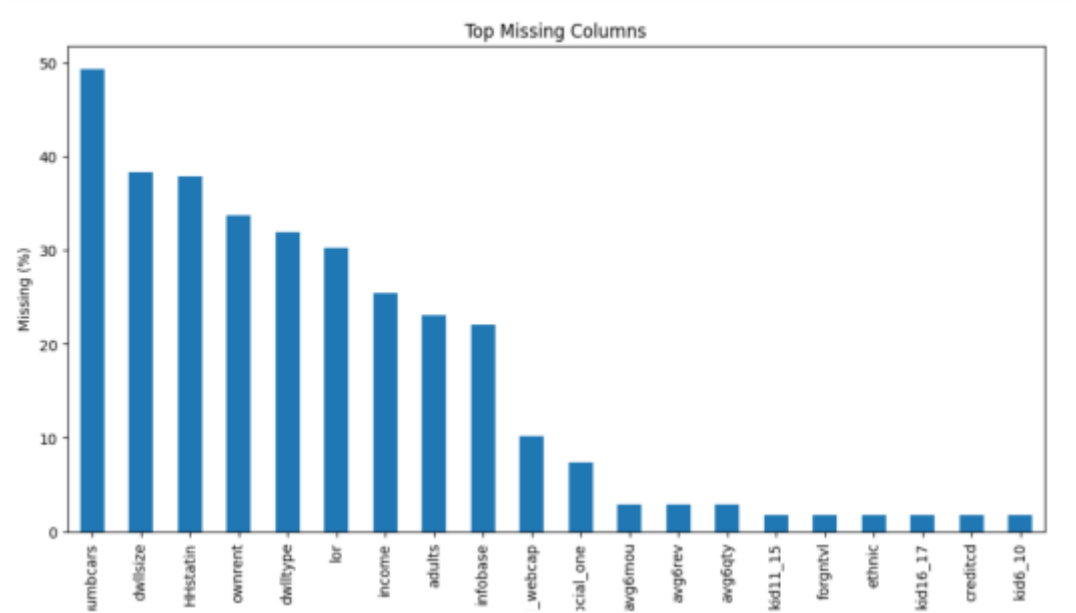
Dataset Overview

- **Dataset Source**
 - Company A Telecom Customer Dataset
 - Two linked tables: **Client.csv** and **Record.csv**
- **Dataset Summary**
 - **Total Customers:** 17,518
 - **Total Features:** 100
 - **Target Variable:** Churn
 - 0 = Customer Stays**
 - 1 = Customer Churns**
- **Data Preparation**
 - Merged both datasets using the customer ID.
 - Handled missing values.
 - Prepared the data for machine learning analysis.

Exploratory Data Analysis

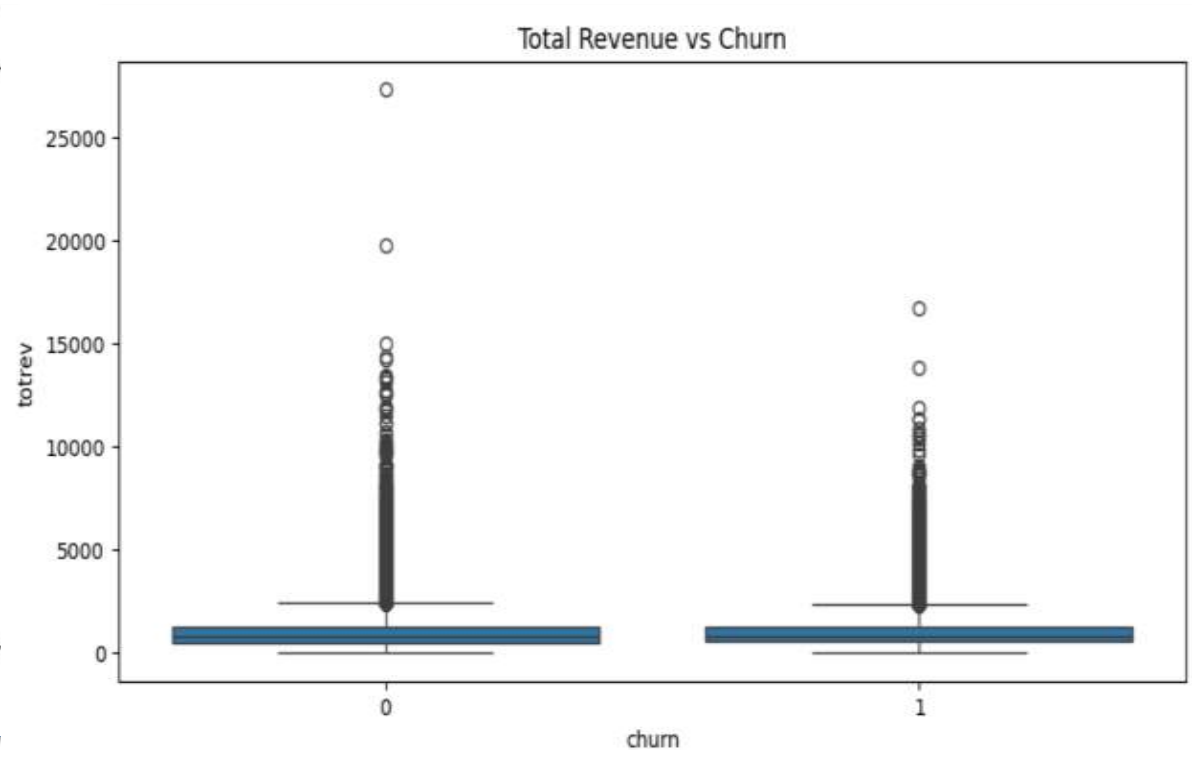
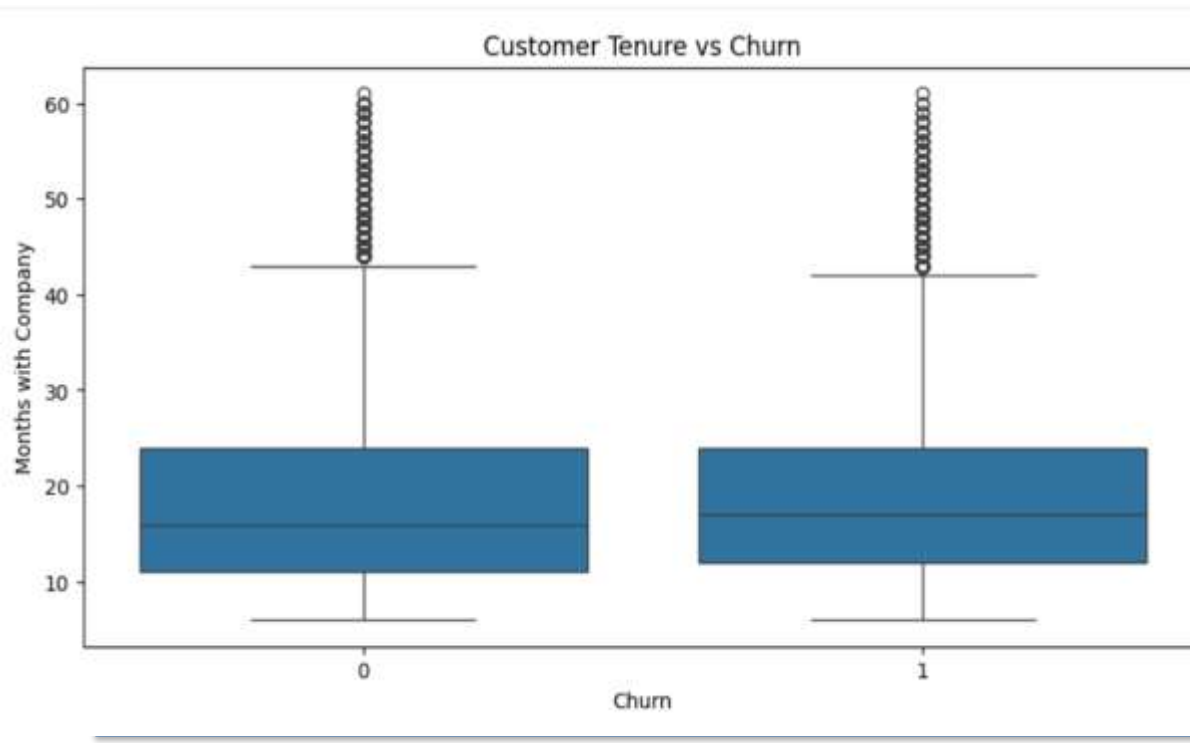
Insert your EDA graphs here:

- Missing Values
- Churn Distribution



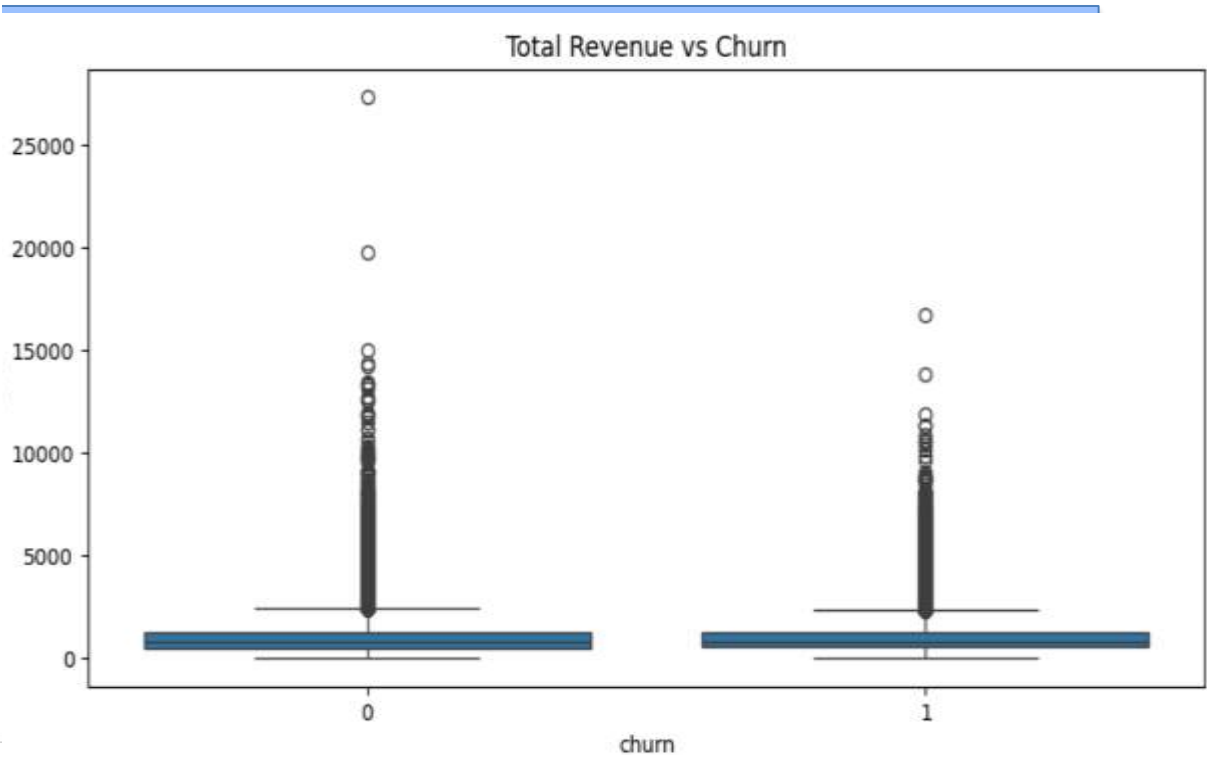
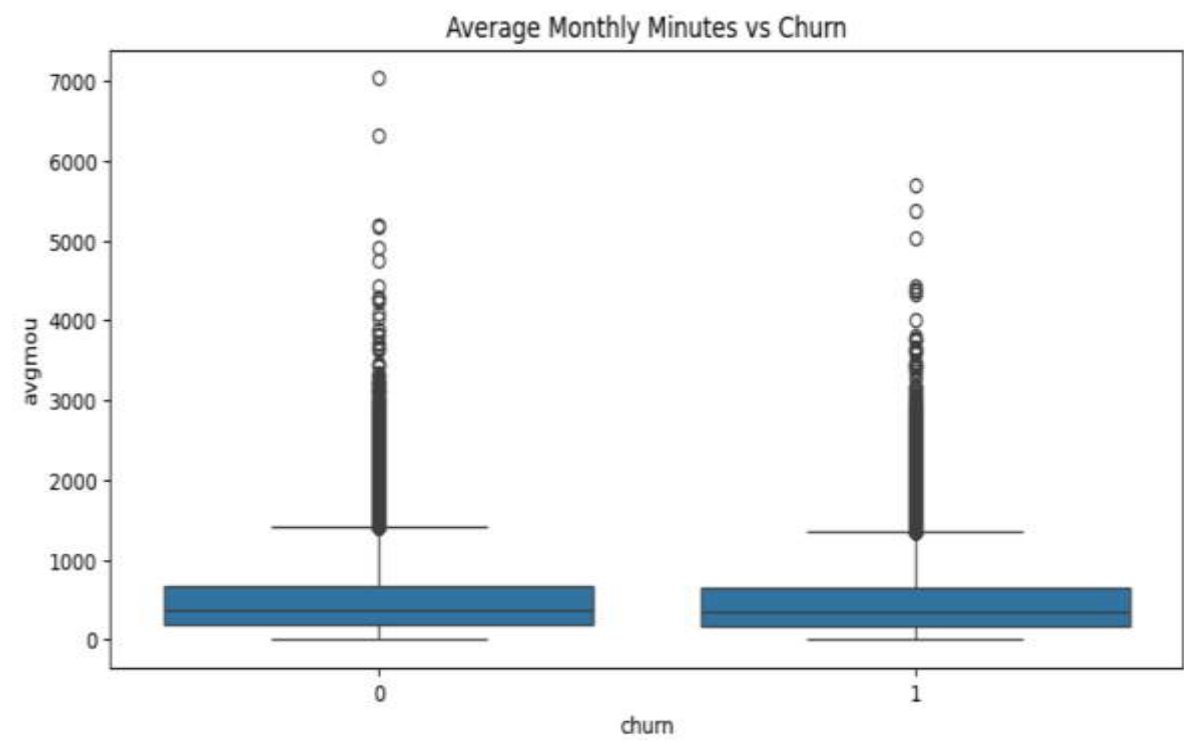
Observation: Replace this text with the observations provided in the chat after inserting the graph.

Customer Behavior Analysis



Observation: Replace this text with the observations provided in the chat after inserting the graph.

Machine Learning Workflow



Observation: Replace this text with the observations provided in the chat after inserting the graph.

Model Performance

Random Forest performed best.

Accuracy: 62.53%

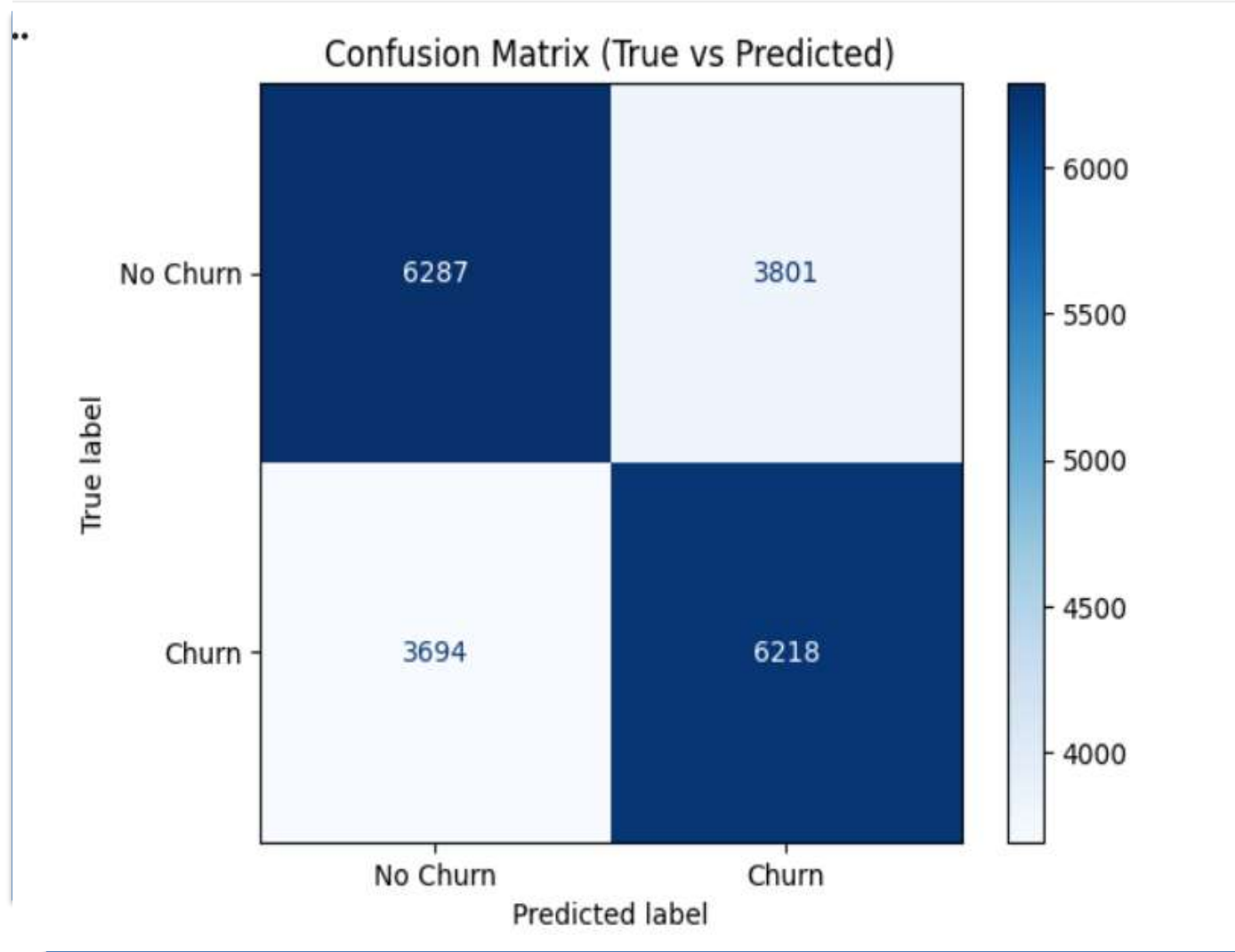
ROC-AUC: 0.675

	Model	Accuracy	Precision	Recall	F1	ROC_AUC
2	Random Forest	0.62525	0.620621	0.627320	0.623953	0.675220
1	Decision Tree	0.59685	0.597676	0.570722	0.583888	0.640529
0	Logistic Regression	0.59755	0.593458	0.596751	0.595100	0.631104

Business Recommendations

- Identify high-risk customers
- Personalized offers
- Loyalty rewards
- Better support
- Increase retention

Confusion Matrix



Explanation:

- Shows True vs Predicted labels.
- Higher diagonal values indicate better classification.
- Supports Random Forest performance.

Conclusion

- Random Forest achieved the best performance.
- Predictive analytics can improve retention.

THANK YOU